A Sharp Preperiod and Explicit Nim-Words for the Four-Move Subtraction Sets $\{3, 7, 9, c\}$

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Preprint v1.0 — 11 November 2026
Not human peer-reviewed

Abstract

For the normal-play one-heap subtraction game with move set $S_c = \{3, 7, 9, c\}$, where $c > 9$ is odd, we determine the complete Sprague–Grundy sequence and its *least* eventual preperiod and period. For $c = 13$ the nim-sequence is purely periodic with least period 16; for every other admissible c its least preperiod is $c + 5$ and its least eventual period is 2. A single block expression gives the full nim-word for all odd $c \geq 23$, and six small values are treated by finite recurrence certificates. This proves, for the slice $a = 3$, an ultimate-periodicity pattern reported by Zhang as a finite computational observation for four-move sets $\{a, 2a + 1, 3a, c\}$. The eventual period-two phenomenon itself is not claimed as new. Historical novelty beyond the compared sources remains unverified.

Status. Research No. 63, preprint v1.0. AI-assisted research. Author-supplied AI-review assessments report MODEL/PROOF/COMPUTATION/SCOPE PASS; a subsequent v0.4 differential review is reported PASS. These reports and author-side primary-source comparisons have distinct scopes. Human peer review and formal proof verification have not been performed. **REFERENCES PARTIAL; NOVELTY INCONCLUSIVE.**

1 Introduction and relationship to earlier work

A finite one-heap subtraction game is specified by a nonempty set of legal move sizes $S \subset \mathbb{Z}_{>0}$. Its Sprague–Grundy sequence is eventually periodic, but determining the entire nim-word along with its *least* preperiod and period can be subtle. Cairns and Ho [2] study ultimate bipartiteness, and Ho [3] studies the expansion of three-element subtraction sets. Zhang [1] studies parameterized move sets and formulates asymptotic-linearity and piecewise-linearity conjectures for their preperiods and periods.

In particular, Zhang's Remark 7 (author-posted revision, p. 10) describes finite calculations for four-move sets

$$S = \{a, 2a + 1, 3a, c\}, \qquad a \geq 3 \text{ odd}, \quad c > 3a \text{ odd},$$

with $3 \leq a \leq 25$ and $c < 500$. For $a = 3$, its exceptional branch $c = 4a + 1$ gives $c = 13$; otherwise the remark reports ultimate period two. The observation alone does not furnish the full all-c nim-word or the least preperiod formula proved here. Theorem 6.3 of the same source treats three-move families; in particular, the earlier candidate $\{3, 5, 8k + 1\}$ is already covered by that theorem and is *not* pursued here. The observed ultimate period-two behavior itself is not claimed as new.

Flammenkamp's dissertation [4] develops explicit four-move nim-word constructions, so this method is prior art. Author-side comparison of the 156-page author-hosted original covered relevant definitions, four-move classifications and constructions, and general results; no direct statement yielding the present full-word and least-preperiod formula was identified in those passages. Its pure-periodicity criterion for the win–loss sequence (Satz 17, printed p. 61) assumes adjacent gaps at most the least move, failing here at $7 - 3 > 3$, and concerns a sequence distinct from the Sprague–Grundy sequence. These bounded checks do not establish historical novelty. The enlarged book edition (ISBN 3-932136-10-1) remains unverified. Larsson and Saha [5] provide survey background.

Among 2026 works, Manabe [6] treats purely periodic *three*-move games, a different parameter regime. We examined Moriwaki [7] at Theorems 2 (p. 8), 3 (p. 10), and 4 (p. 23). Theorem 2 assumes, for the entire sorted move set $s_1 < \cdots < s_r$, the adjacent-gap bounds $s_{j+1} \leq s_j + s_1$. Our set violates this at $7 > 3 + 3$. Theorem 4 has a different structure: it starts with a *base* set $B = \{b_1 < \cdots < b_r\}$ satisfying

these bounds and constructs $B_l = B \cup \{b_1 + b_r, 2b_1 + b_r, \ldots, lb_1 + b_r\}$ for $l \geq 0$. It does not directly yield our $S_c = \{3, 7, 9, c\}$ for any base B and l: if $l = 0$, the base is S_c and its gap bound fails; if $l \geq 1$, every element of $B \subseteq S_c$ is odd, so the required added move $b_1 + b_r$ is even and cannot belong to S_c. Theorem 3 treats three moves. Thus none of these specified theorems directly covers the present four-move formula; this does not exclude indirect consequences of other work. Inoue and Inazu [8] study subtraction of suffixes from *words*. A subsequent author-side targeted reading of their HTML full text, including Theorem 2.9 on regularity of the language of winning positions and the Grundy-number discussion, identified no direct theorem giving the present integer-heap formula. This is a bounded source comparison, not an exclusion of all indirect implications. No exact predecessor for our formula was identified in these bounded checks, but **NOVELTY remains INCONCLUSIVE**.

2 Definitions and statement

For $n \geq 0$, let $g_S(n)$ be defined by

$$g_S(n) = \operatorname{mex}\{g_S(n - s) : s \in S,\ s \leq n\}, \qquad \operatorname{mex} \emptyset = 0. \tag{1}$$

The *least eventual period* p is the least $p \geq 1$ for which there exists $T \geq 0$ such that $g_S(n + p) = g_S(n)$ for all $n \geq T$. The *least preperiod* T is the least starting point of a periodic suffix. These minima are considered for the same eventually periodic sequence.

For a finite digit word U, write U^r for r concatenated copies, U^∞ for its infinite repetition, and juxtaposition for concatenation. Set

$$W = \mathtt{000111022133020}, \qquad |W| = 15.$$

We use $W_{<11} = \mathtt{00011102213}$ for the first 11 symbols of W.

Theorem 1. *Let c be any odd integer greater than 9, and let $S_c = \{3, 7, 9, c\}$. The complete Sprague–Grundy word $\mathcal{G}_c = (g_{S_c}(0), g_{S_c}(1), \ldots)$ is as follows.*

(i) For every odd $c \geq 23$,

$$\mathcal{G}_c = W\,(\mathtt{10})^{(c-15)/2}\,\mathtt{12103}\,(\mathtt{01})^\infty. \tag{2}$$

(ii) For the remaining odd values $c > 9$,

$$\begin{aligned}
\mathcal{G}_{11} &= W_{<11}\,\mathtt{32203}\,(\mathtt{01})^\infty,\\
\mathcal{G}_{13} &= (\mathtt{0001110221330241})^\infty,\\
\mathcal{G}_{15} &= W\,\mathtt{13102}\,(\mathtt{01})^\infty,\\
\mathcal{G}_{17} &= W\,\mathtt{10\,12103}\,(\mathtt{01})^\infty,\\
\mathcal{G}_{19} &= W\,(\mathtt{10})^2\,\mathtt{14102}\,(\mathtt{01})^\infty,\\
\mathcal{G}_{21} &= W\,(\mathtt{10})^3\,\mathtt{13102}\,(\mathtt{01})^\infty.
\end{aligned}$$

The least *preperiod and least eventual period are*

$$(T, p) = \begin{cases} (0, 16), & c = 13,\\ (c + 5, 2), & \textit{otherwise.} \end{cases} \tag{3}$$

3 Proof for the infinite family $c \geq 23$

Lemma 1. *For the three-move set $S_0 = \{3, 7, 9\}$, one has*

$$\mathcal{G}_{S_0} = W\,(\mathtt{10})^\infty.$$

Proof. Direct evaluation of (1) for $0 \leq n \leq 23$ gives the stated first 24 digits. For $n \geq 24$, the three predecessors $n - 3$, $n - 7$, $n - 9$ are all at least 15 and have parity opposite to n. In the proposed tail, an even index has value 0 and an odd index value 1. Hence the three predecessor values are all 1 if n is even, and all 0 if n is odd. Their mex is respectively 0 or 1, proving the recurrence and the displayed word by induction. □

Proof of Theorem 1(i). Fix odd $c \geq 23$, and let f denote the right-hand side of (2). For all $0 \leq n < c$, move c is illegal, and f agrees with the base word in Lemma 1. The five new symbols at indices $c, c+1, \ldots, c+4$ are `12103`. The predecessor symbols for moves $3, 7, 9, c$ give the following exact mex checks:

n	predecessor values	mex
c	$(0,0,0,0)$	1
$c+1$	$(1,1,1,0)$	2
$c+2$	$(0,0,0,0)$	1
$c+3$	$(1,1,1,1)$	0
$c+4$	$(2,0,0,1)$	3

These evaluations use $c - 9 \geq 14$ and the explicitly checked terminal digit of W at index 14; the stable alternating base phase applies thereafter.

The candidate f has the following properties, all read directly from its displayed word:

(A) $f(j) \neq 0$ for every odd $j \geq 5$;

(B) $f(j) \neq 1$ for every even $j \geq 6$;

(C) $f(j) = 0$ for every even $j \geq c - 3$, except that $f(c+1) = 2$.

We verify the recurrence for all $n \geq c+5$. If n is even, all four legal predecessor indices $n-3, n-7, n-9, n-c$ are odd and at least 5. By (A), none of their values is 0, hence the mex is $0 = f(n)$.

If n is odd, these four predecessor indices are even, with $n - c \geq 6$. By (B), none has value 1. The three distinct indices $n - 3, n - 7, n - 9$ all lie at or beyond $c - 3$. By (C), at most one of their values is nonzero, so at least two equal 0. The mex is consequently $1 = f(n)$. Strong induction now shows that the candidate word satisfies (1), and uniqueness of the recursion yields $f = \mathcal{G}_c$.

The alternating tail begins at $T = c + 5$, an even index, and contains both 0 and 1, so its least period is 2. Also $f(T-1) = f(c+4) = 3$ whereas $f(n) \in \{0, 1\}$ for every $n \geq T$. If any period $q \geq 1$ held from $T' < T$, it would imply $f(T-1) = f(T-1+q)$. But $T - 1 + q \geq T$, so the right-hand side is 0 or 1, a contradiction. Thus *no period of any length* can begin earlier, establishing the least preperiod $T = c + 5$. □

4 The six small parameters: finite proof certificates

For $c \in \{11, 13, 15, 17, 19, 21\}$, the words in Theorem 1(ii) are specified by a finite prefix and a periodic suffix. The following principle makes the remaining recurrence verification finite.

Lemma 2 (Finite periodic-word check)**.** *Let $c = \max S$ and let a proposed word f be periodic from index T with period p. If f satisfies* (1) *for every $0 \leq n \leq T + p + c - 1$, then f satisfies the recurrence for every $n \geq 0$.*

Proof. Proceed by strong induction. For $n \geq T + p + c$, all predecessors for both n and $n - p$ are legal, and $(n - p) - s \geq T$ for every $s \in S$. Periodicity of the candidate word makes the lists of predecessor values at n and $n - p$ equal. Thus their mex values agree, and since $f(n) = f(n - p)$, the recurrence at $n - p$ (already established at a smaller index) implies the recurrence at n. □

c	candidate T	candidate p	last checked index $T + p + c - 1$	status
11	16	2	28	PASS
13	0	16	28	PASS
15	20	2	36	PASS
17	22	2	40	PASS
19	24	2	44	PASS
21	26	2	48	PASS

The accompanying code checks each complete proposed word against the mex recursion throughout its finite certificate range; Lemma 2 upgrades this to an infinite identity. For $c = 13$, the repeated 16-letter word has no periods dividing 16 other than 16 itself, and being purely periodic it has least preperiod zero. For each of the other five cases, the symbol immediately before $T = c + 5$ lies outside $\{0, 1\}$, whereas all

symbols at and after T lie inside $\{0,1\}$. The arbitrary-period contradiction in the proof above gives sharp minimality of T; the nonconstant alternating tail has least period two.

The finite certificates are an essential computational component of the proof, not a claim of exhaustive search over all subtraction sets. Their sources are included unchanged from the audited REV3 package.

5 Reproducibility and limitations

The audit package includes (i) a Python verifier for the finite candidate words; (ii) a separately coded C++17 verifier for odd $c \geq 23$; and (iii) a C++17 verifier for six exceptional parameter values. Their frozen SHA256 values and invocation instructions appear in the accompanying manifest and README. The author-provided Windows PowerShell 5.1 record reports all finite checks passed; the archive contains only a privacy-safe summary, not the raw user-machine log. An author-supplied AI reviewer also reports an independently written recurrence enumeration over 996 odd parameters $11 \leq c \leq 2001$, agreeing with all formulas. Such checks support reproducibility but cannot alone establish an infinite theorem; the proof above supplies that argument.

Historical novelty is not certified. The observational period-two claim is already present in Zhang's Remark 7, whereas the prospective contribution here is the all-odd-c proof for $a = 3$, sharp preperiods, and complete nim-words. The author-hosted original of Flammenkamp's dissertation was obtained, authenticated by SHA256, converted to a 156-page PDF, and its relevant passages compared directly; no direct anticipation was identified *in those examined passages*. The inspected Zhang source is the author's revision dated 30 July 2025, not a guaranteed byte-identical journal edition. Moriwaki's Theorem 2 fails its full-set gap hypothesis here, and Theorem 4 cannot construct our all-odd set from an admissible base (the first adjoined move would be even); this is not a comprehensive theorem-implication survey. Inoue–Inazu was subsequently compared by author-side targeted full-text reading; no direct predecessor was identified within that check. The enlarged Flammenkamp book remains unverified. Accordingly, **REFERENCES PARTIAL; NOVELTY INCONCLUSIVE**. This is a limited, publicly released mathematical preprint, not human peer-reviewed. No claim is made for even c, general $a > 3$, or arbitrary subtraction sets. The unchanged verifiers were rerun during v1.0 preparation under Linux and reproduced the recorded finite-check counts; a new Windows v1.0 execution was not performed.

Rights and disclosure. Copyright © 2026 Ryutaro Yonezu. All rights reserved. This research was conducted with AI assistance and computational verification. Human peer review: not performed. Publication type: Preprint; version: v1.0.

References

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